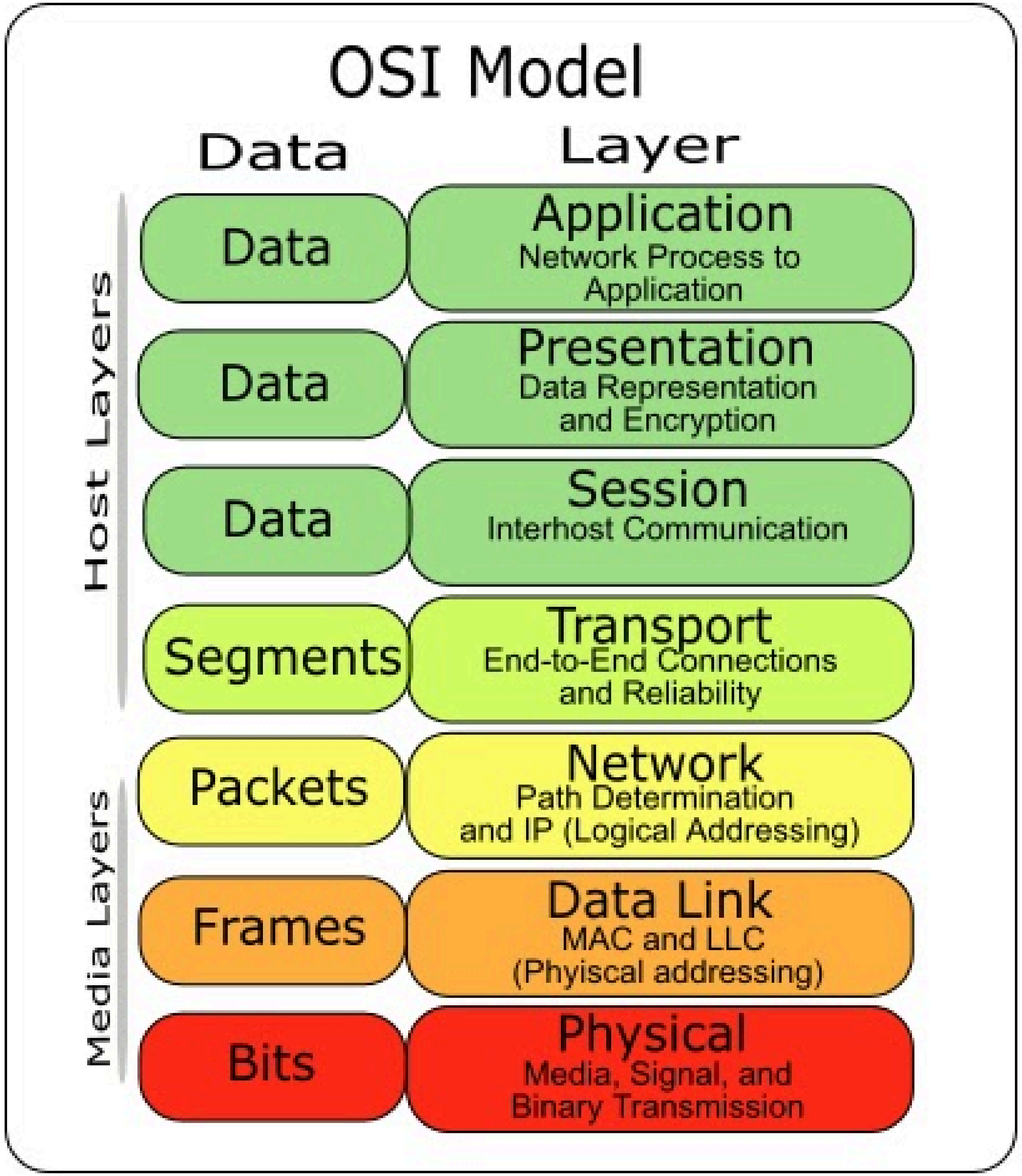
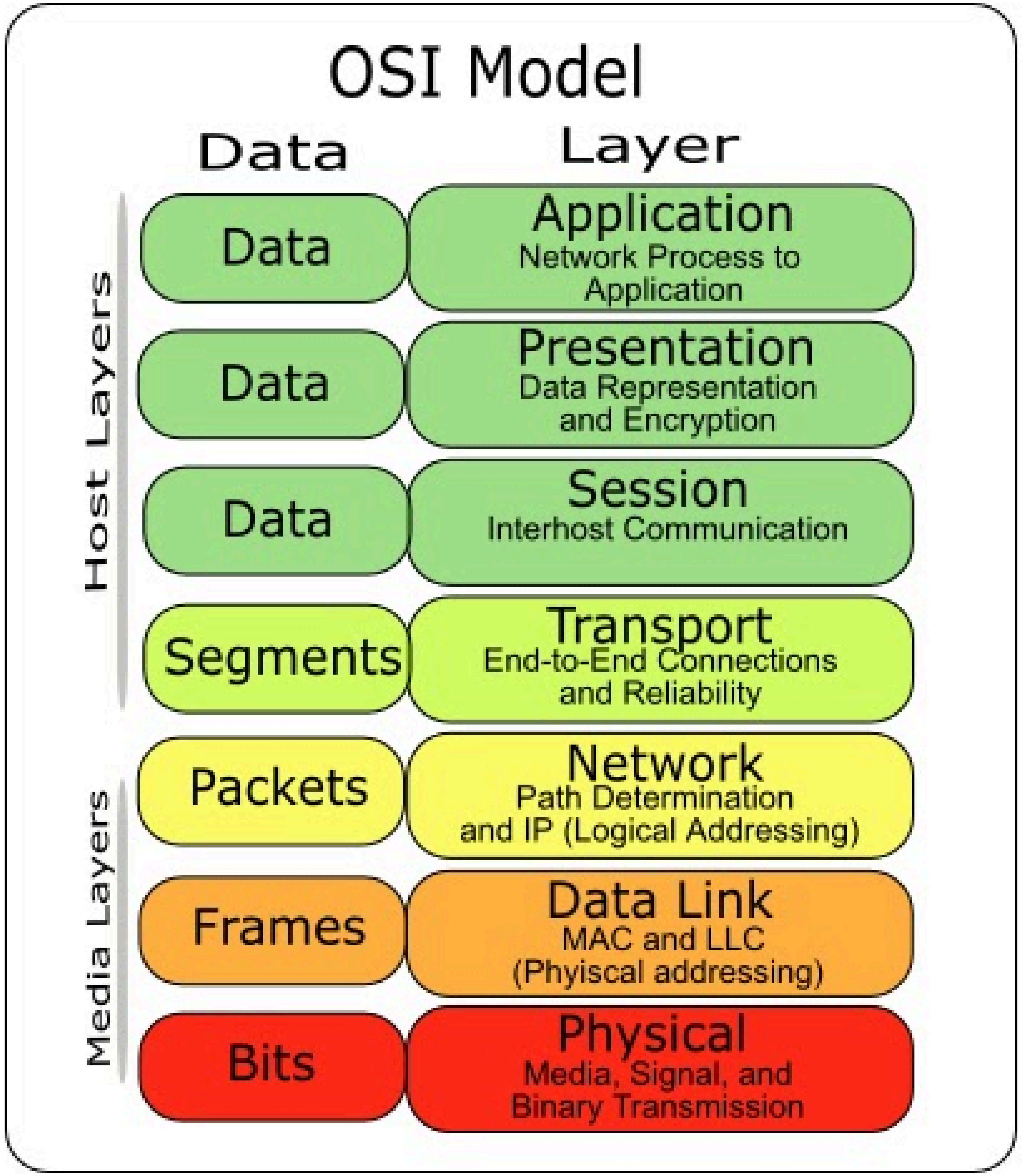


Which layer does **Router** operate on?

Which layer does **Switch** operate on?

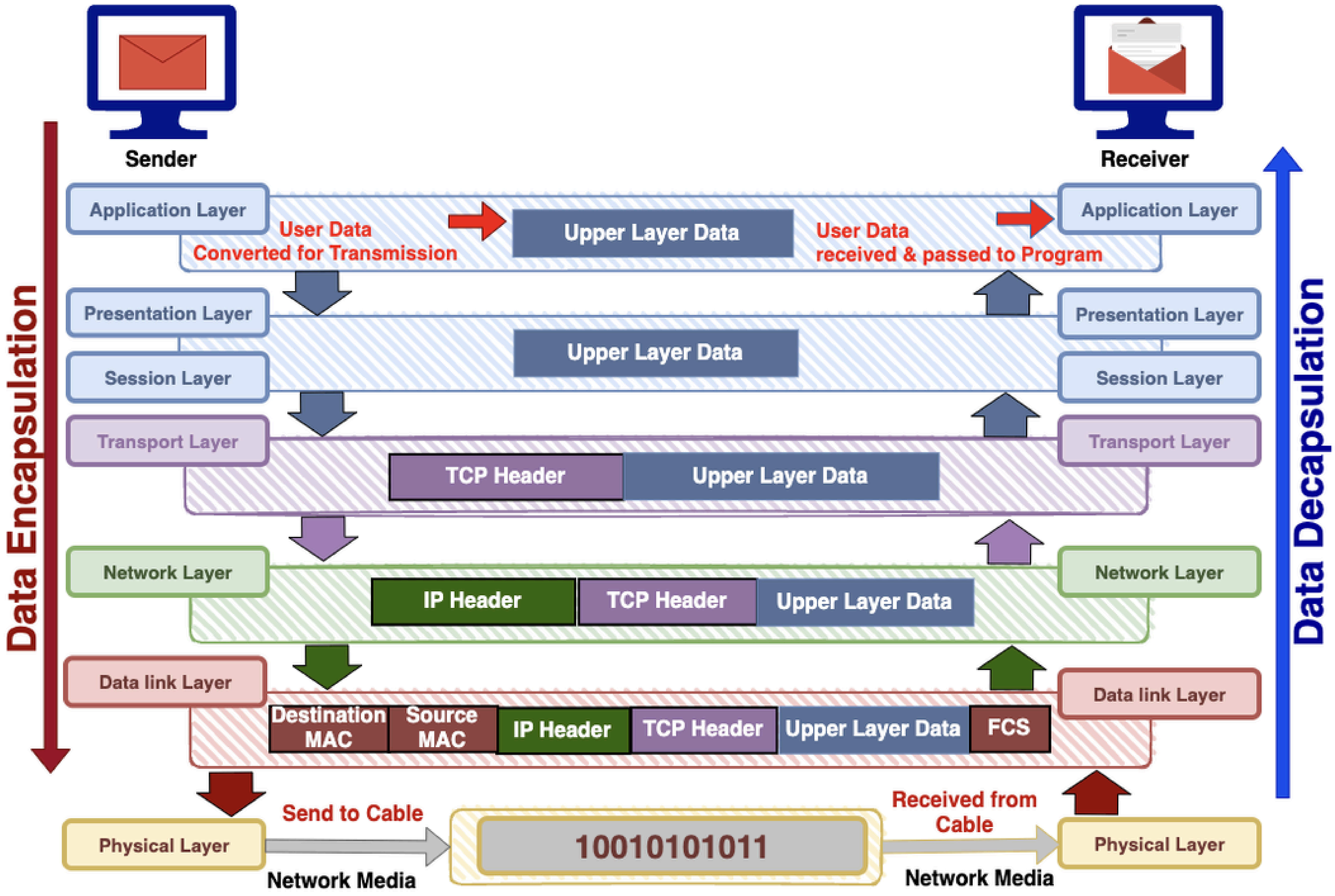


Which layer does **Router** operate on?

Layer 3 (Network)

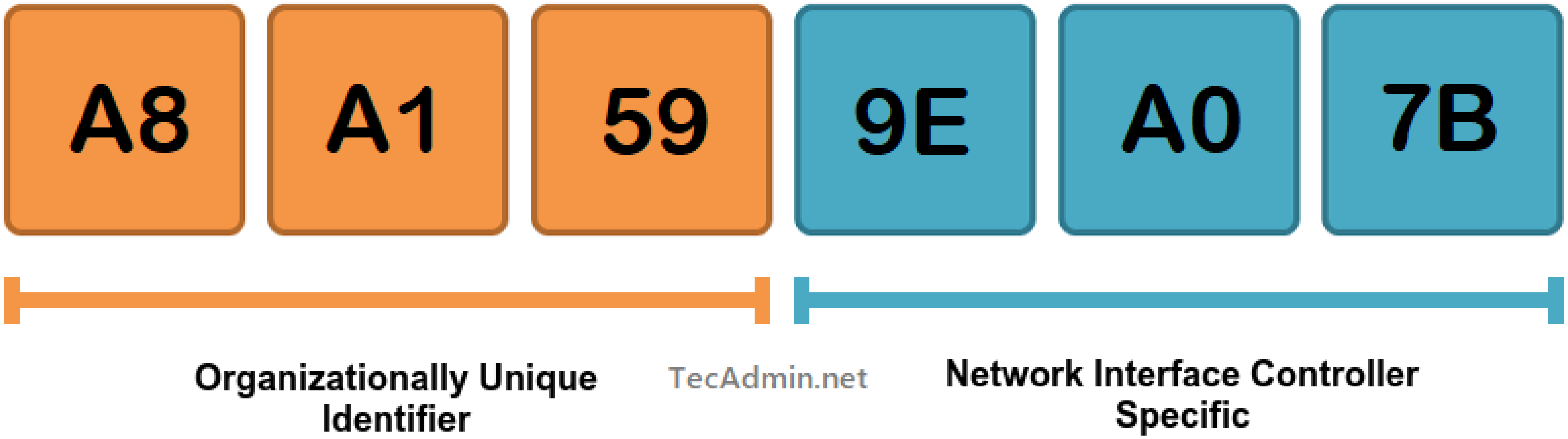
Which layer does **Switch** operate on?

Layer 2 (Data Link)

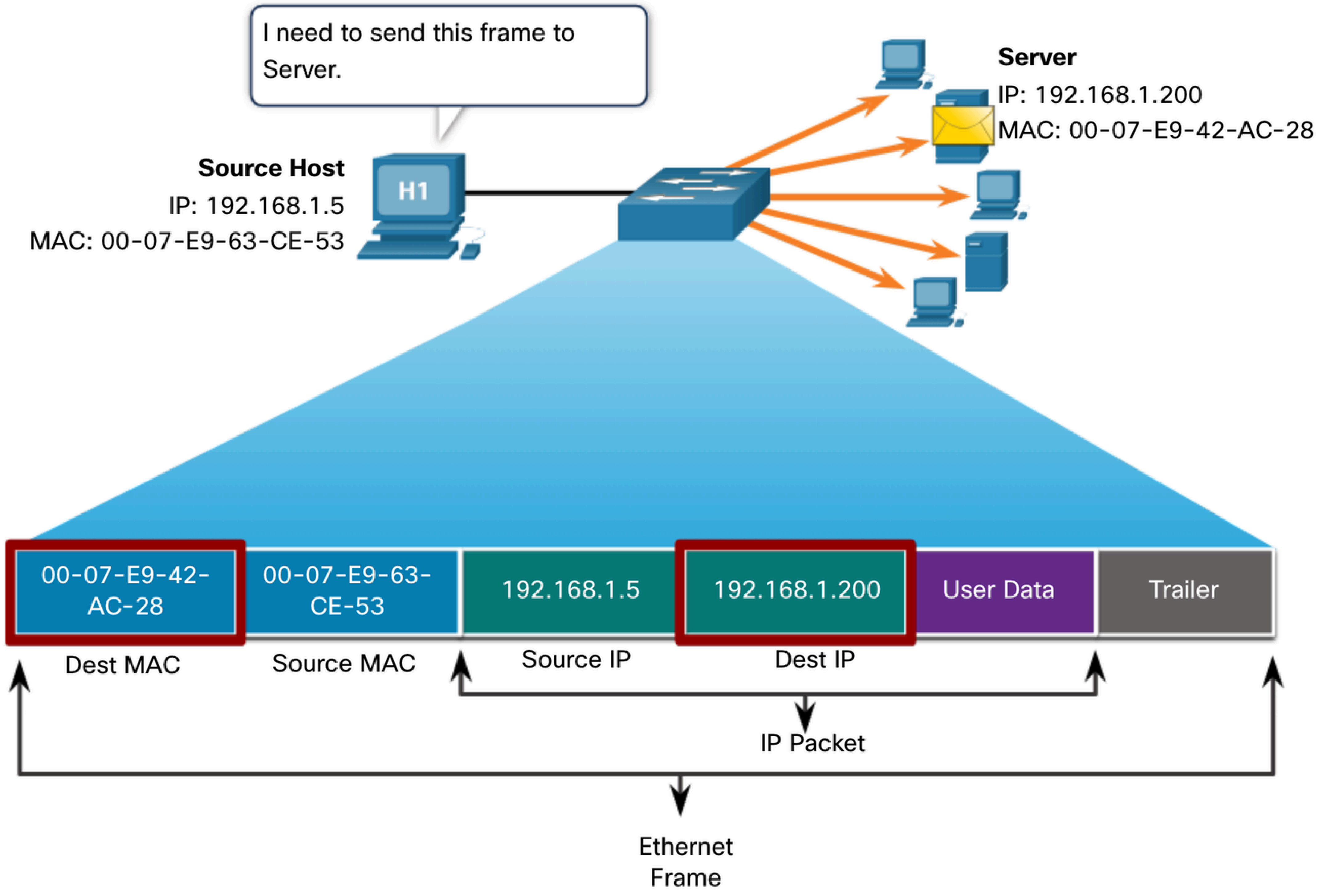

Sender
Receiver
Data Encapsulation
Data Decapsulation
Application Layer
User Data
Converted for Transmission
Upper Layer Data
User Data
received & passed to Program
Application Layer
Presentation Layer
Session Layer
Upper Layer Data
Presentation Layer
Session Layer
Transport Layer
TCP Header
Upper Layer Data
Transport Layer
Network Layer
IP Header
TCP Header
Upper Layer Data
Network Layer
Data link Layer
Destination MAC
Source MAC
IP Header
TCP Header
Upper Layer Data
FCS
Data link Layer
Physical Layer
Send to Cable
10010101011
Received from Cable
Physical Layer
Network Media
Network Media

MAC

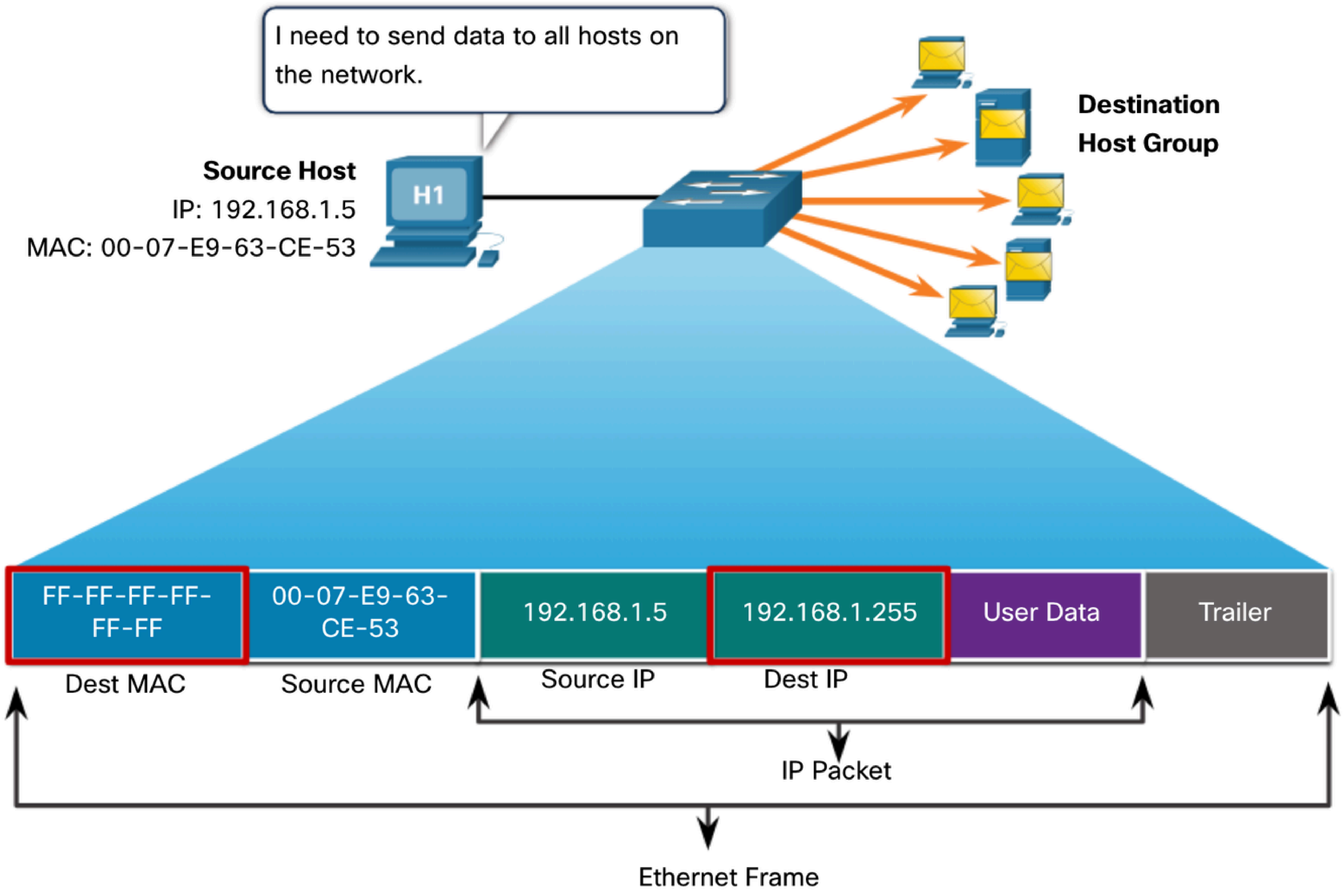
Media Access Control Address



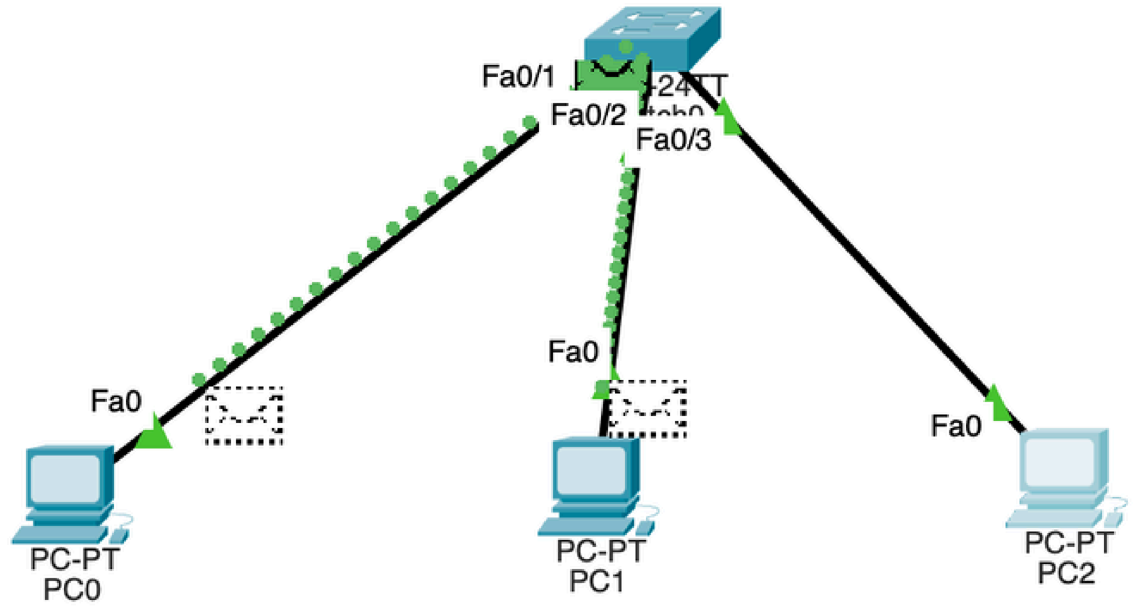
Unicast MAC-address

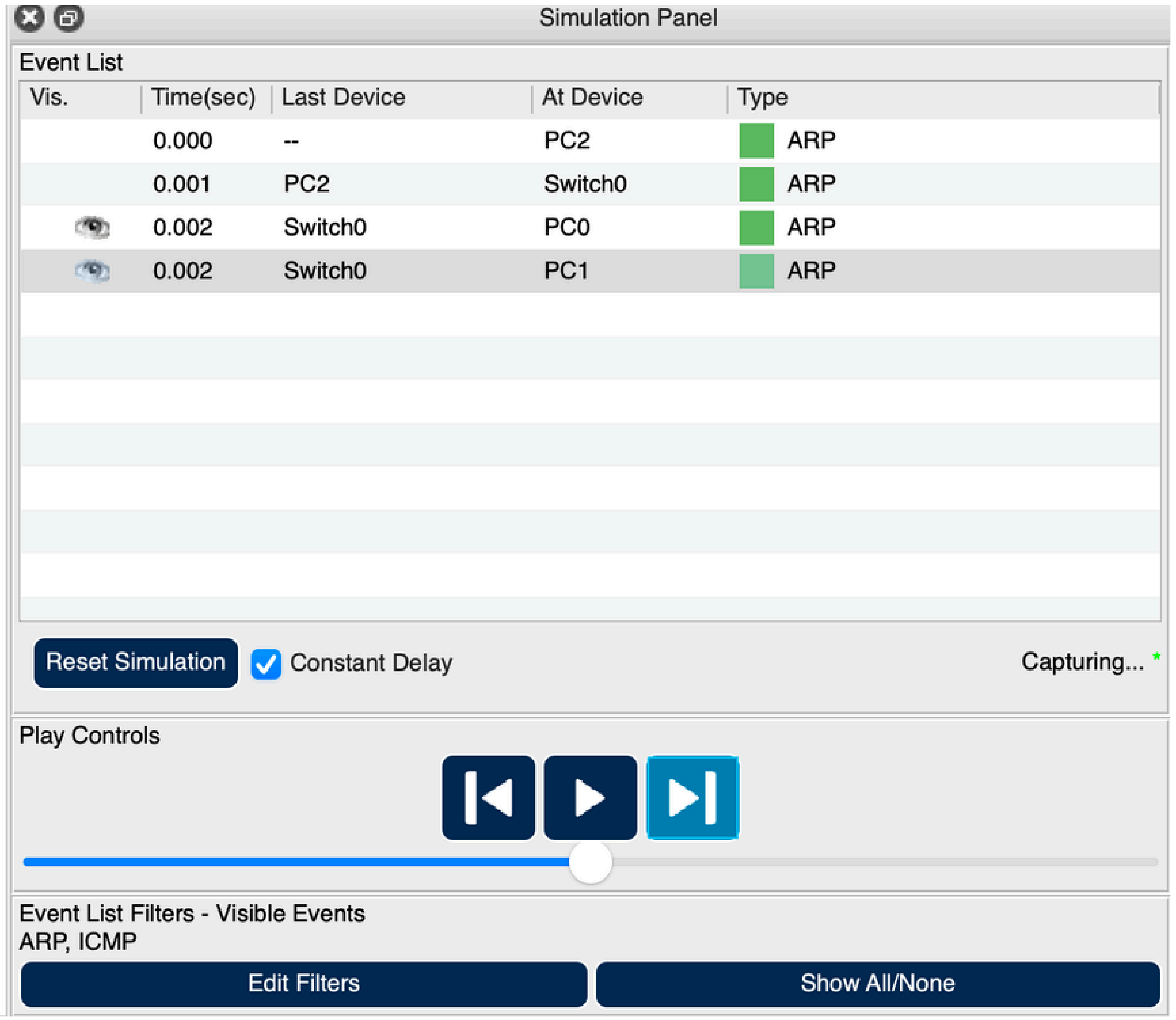


Broadcast MAC-address

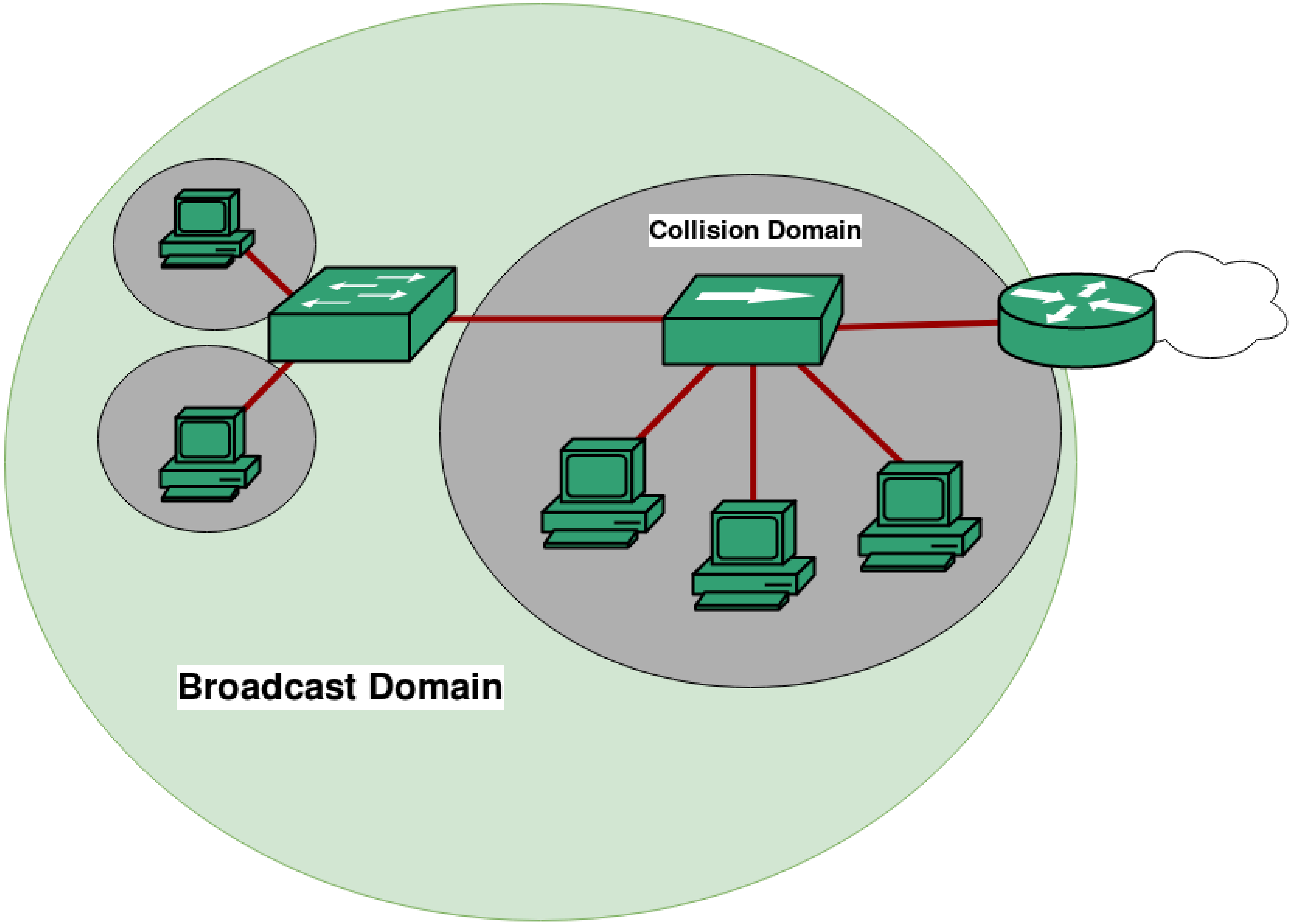


ARP request broadcasting





Broadcast and Collision Domains

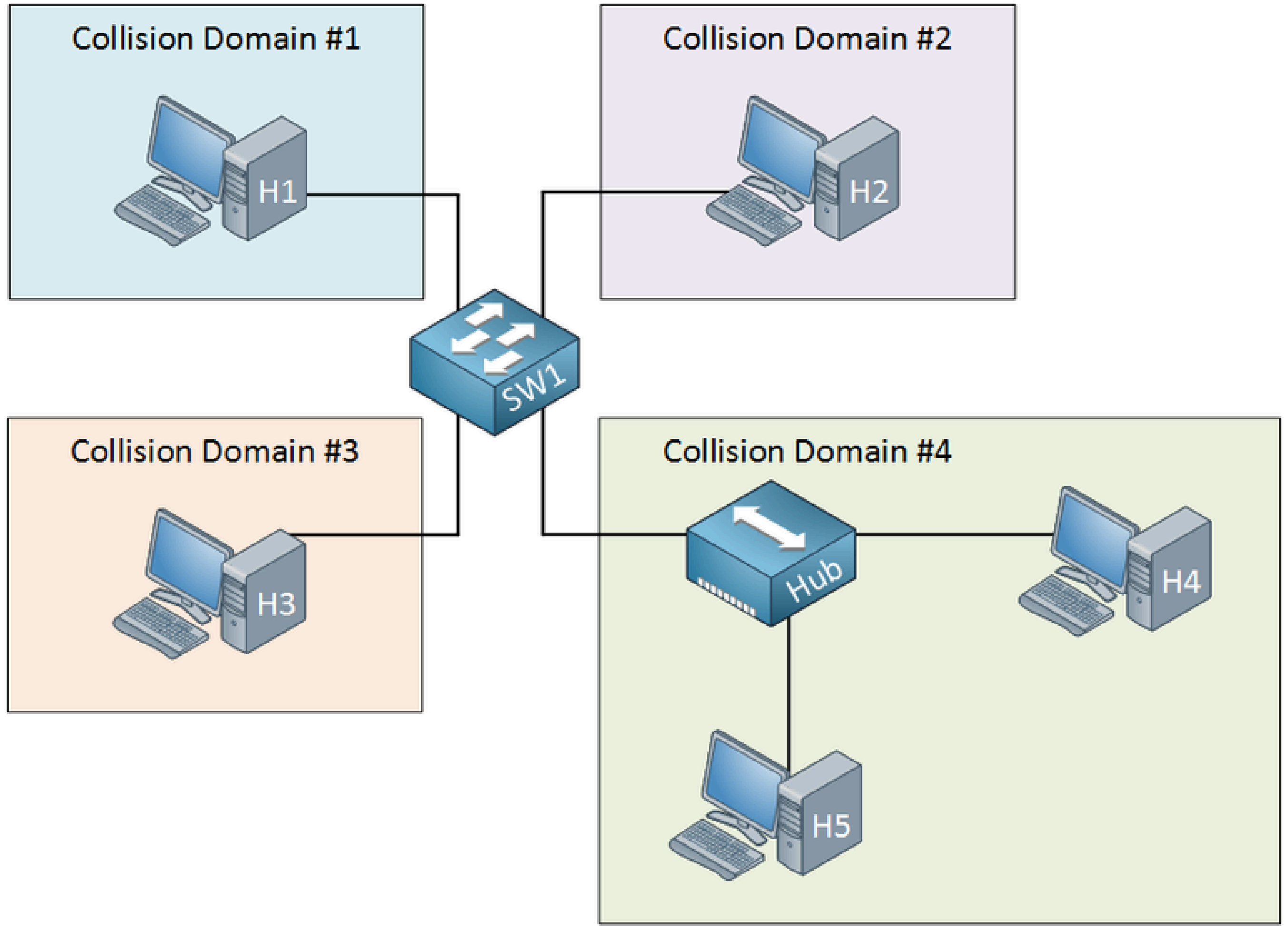


Collision domain

A collision domain is a network segment where data packets can “collide” when two devices send frames at the same time

Collisions occur only in **half-duplex** networks (when using hubs)

Switches break collision domains - each port on a switch is its own collision domain

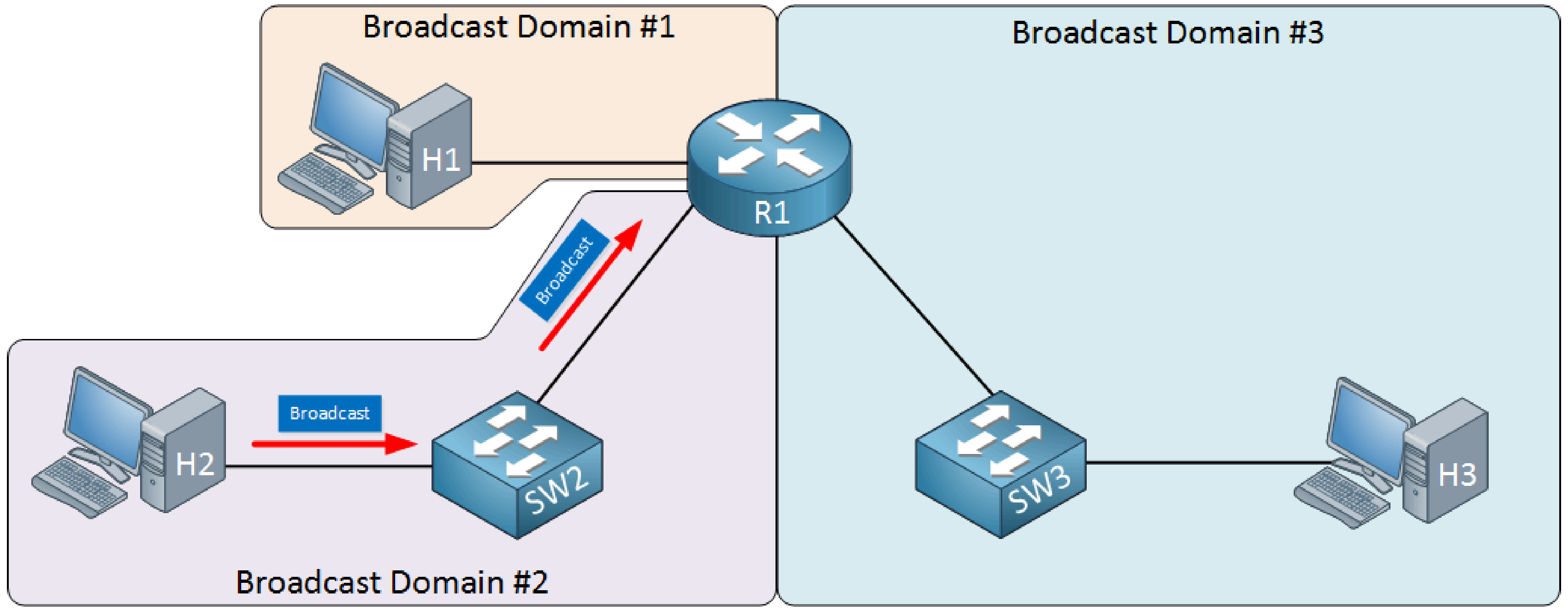


Broadcast domain

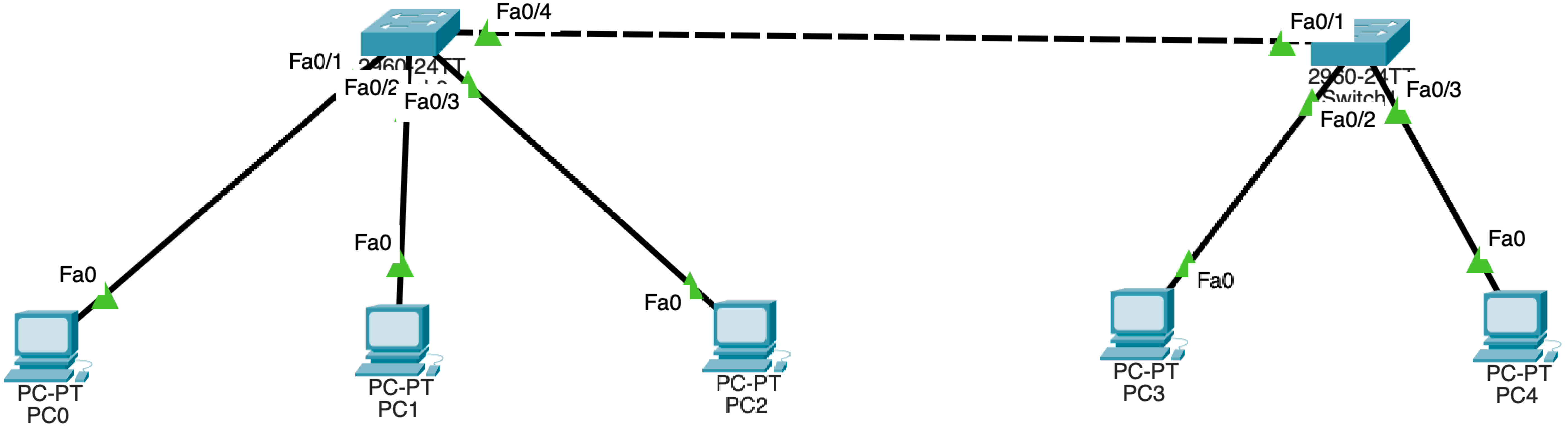
A broadcast domain is a logical area in which any **broadcast frame** (for example, ARP request) is forwarded to all devices

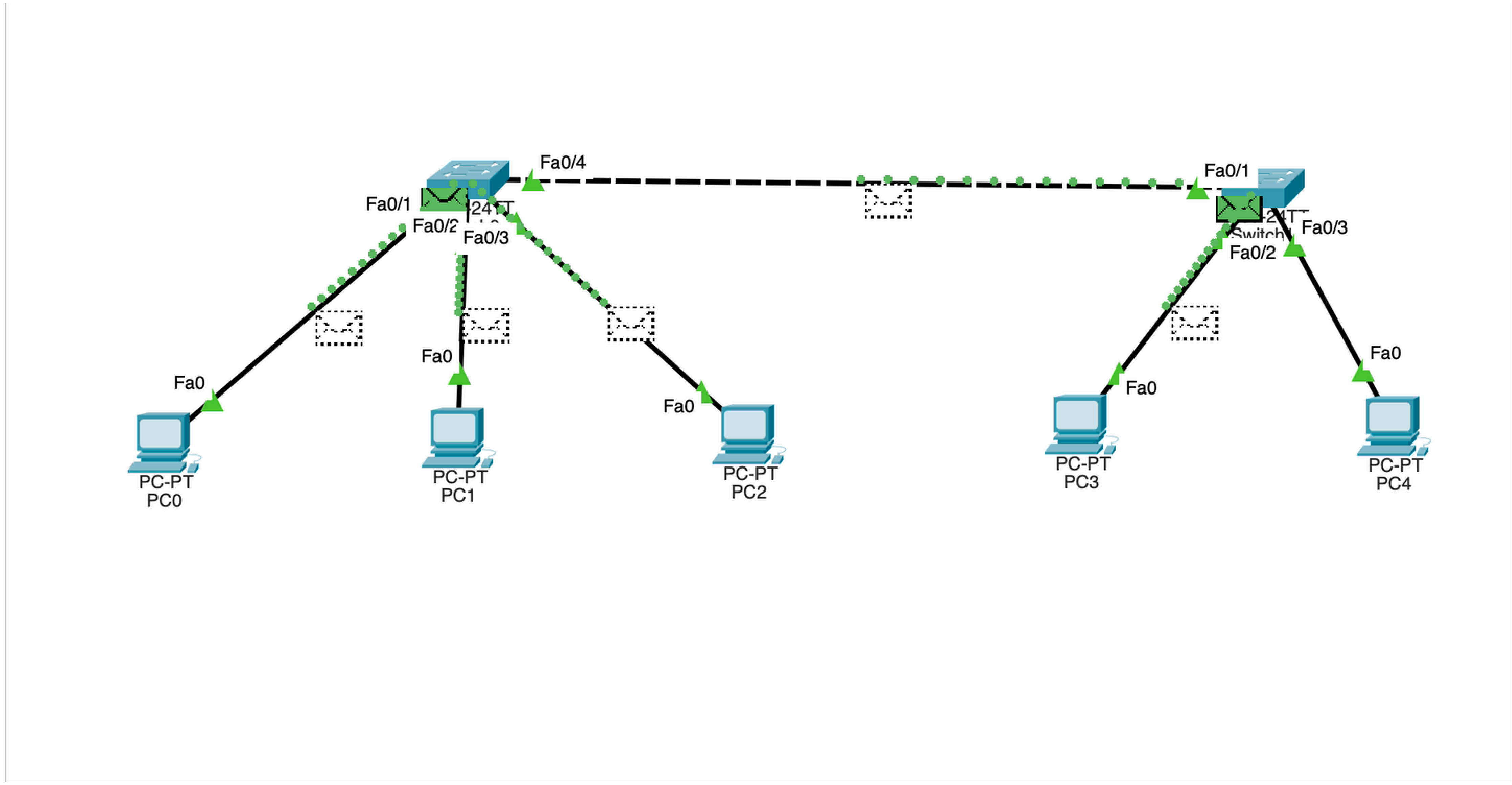
Broadcasts are limited by **routers** (Layer 3 devices)

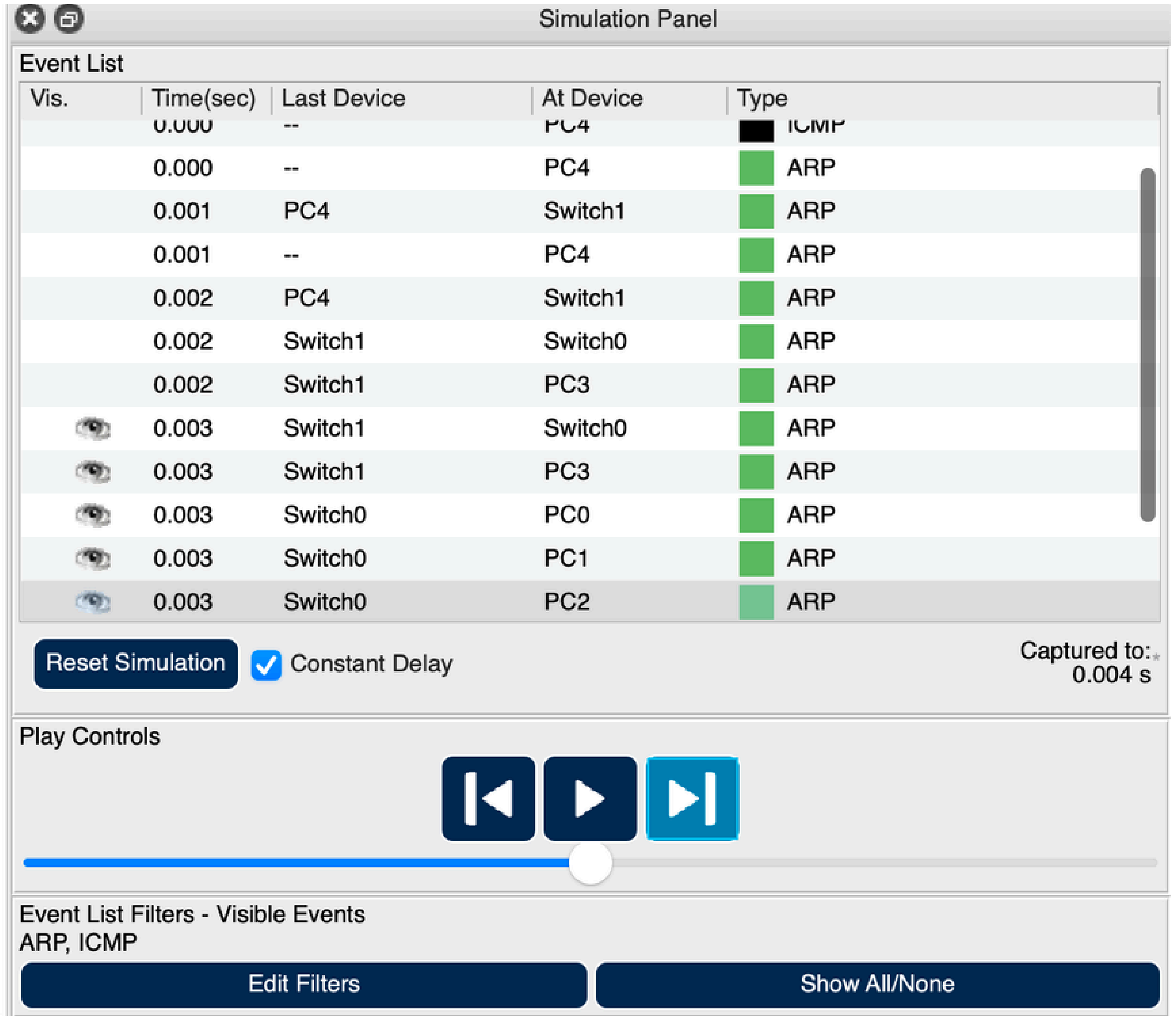
Switches do not block broadcasts **by default**



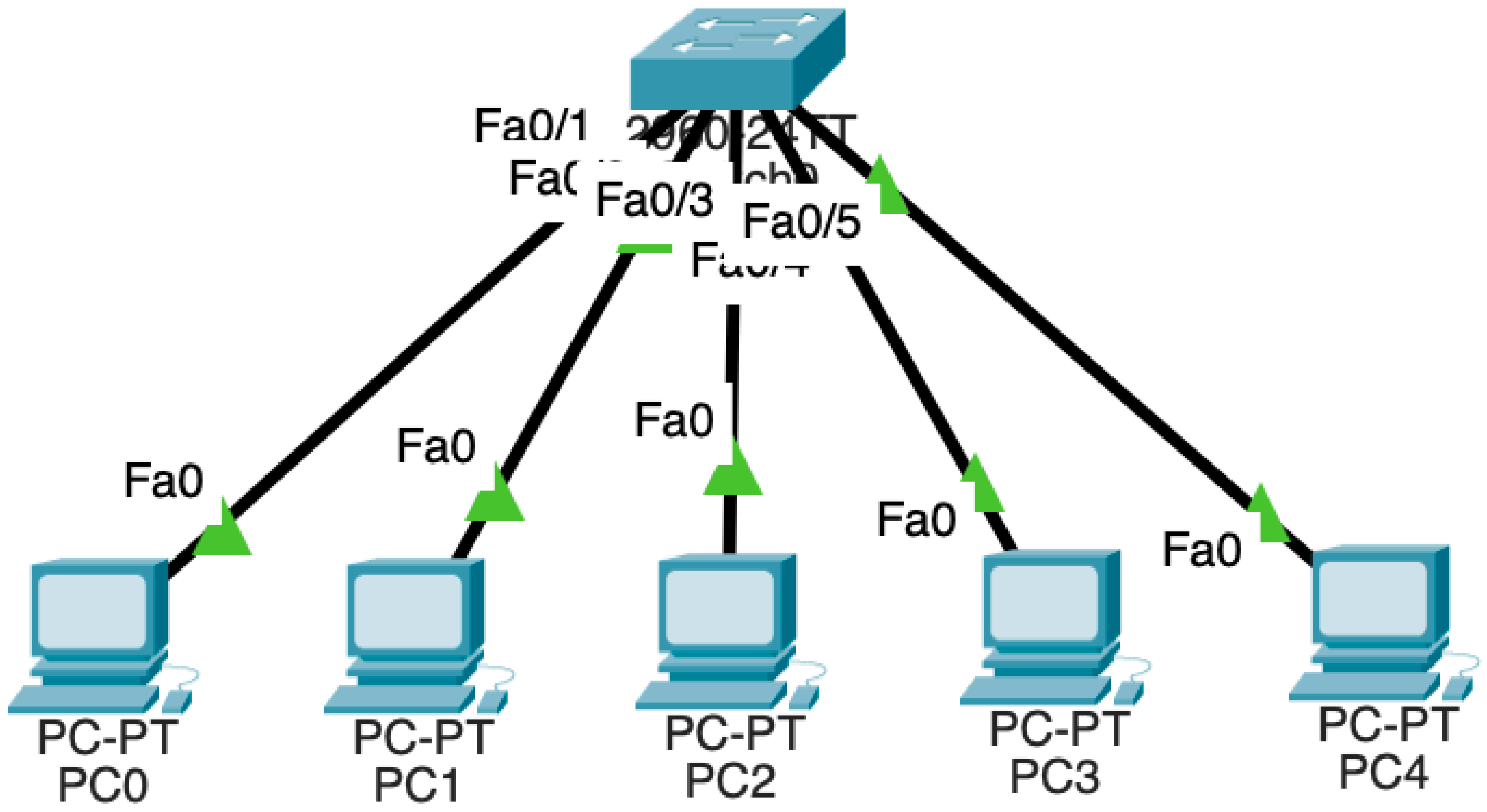
What is the broadcast domain here?




Fa0/4
Fa0/1
Fa0/2
Fa0/3
Fa0/1
Fa0/3
Fa0/2
Fa0
Fa0
Fa0
Fa0
Fa0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4

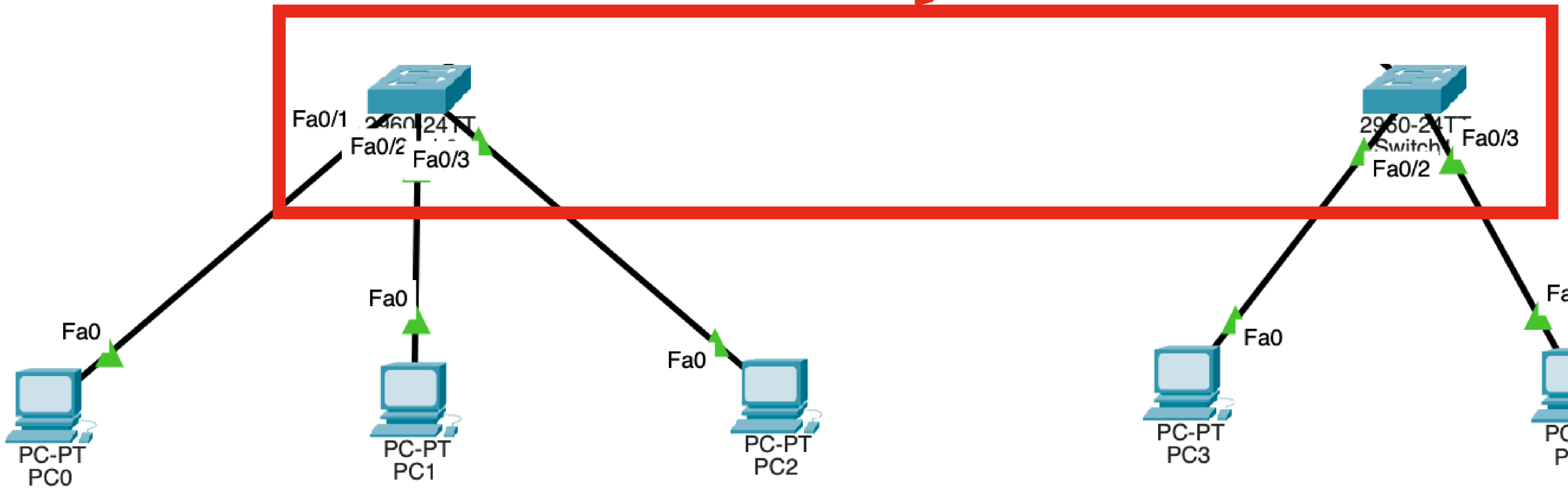

Simulation Panel
Event List
Vis. Time(sec) Last Device At Device Type
0.000 -- PC4 ICMP
0.000 -- PC4 ARP
0.001 PC4 Switch1 ARP
0.001 -- PC4 ARP
0.002 PC4 Switch1 ARP
0.002 Switch1 Switch0 ARP
0.002 Switch1 PC3 ARP
0.003 Switch1 Switch0 ARP
0.003 Switch1 PC3 ARP
0.003 Switch0 PC0 ARP
0.003 Switch0 PC1 ARP
0.003 Switch0 PC2 ARP
Reset Simulation
Constant Delay
Captured to:
0.004 s
Play Controls
Event List Filters - Visible Events
ARP, ICMP
Edit Filters
Show All/None

How to divide the network?



What is the solution?

Too expensive

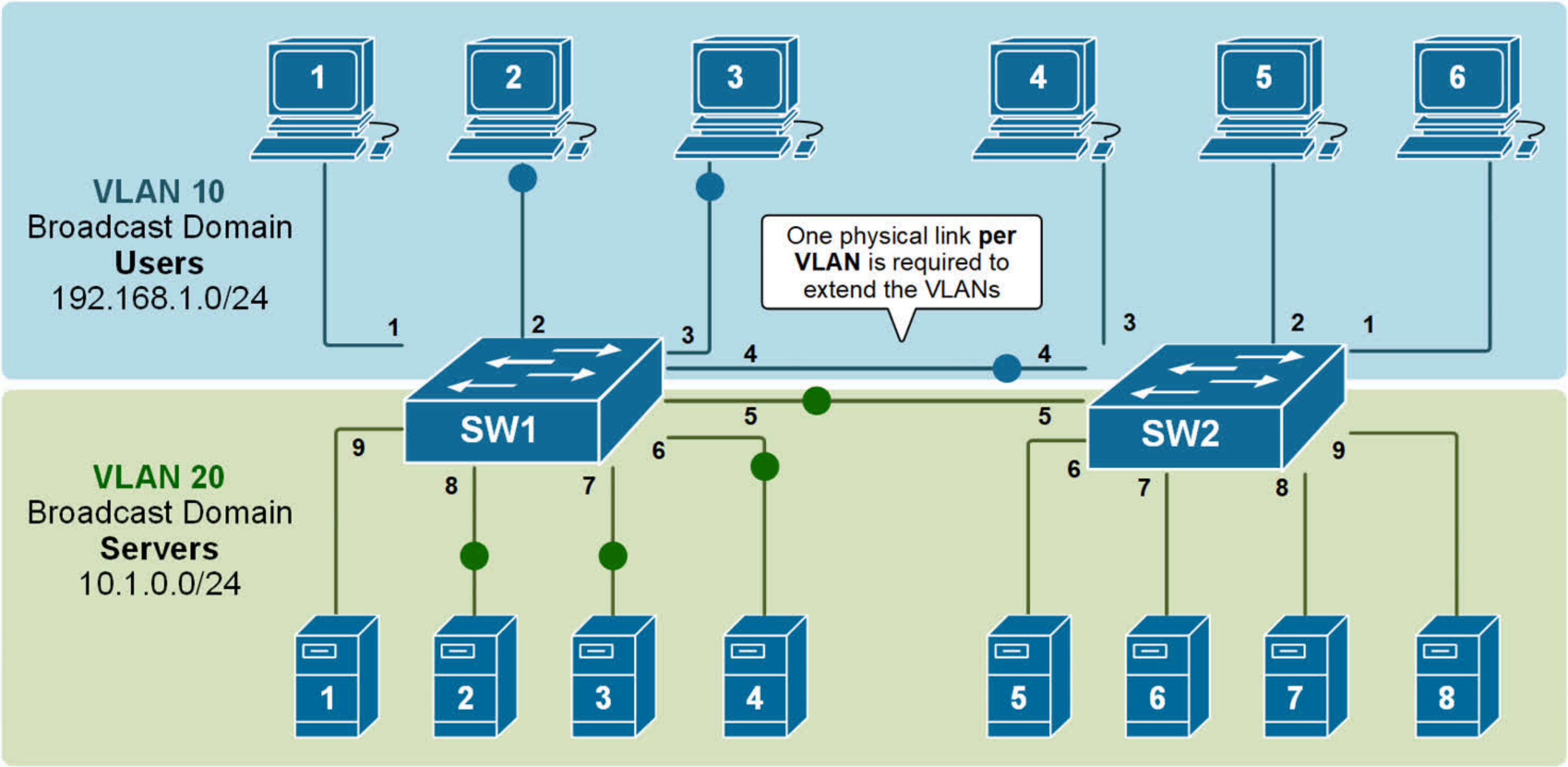


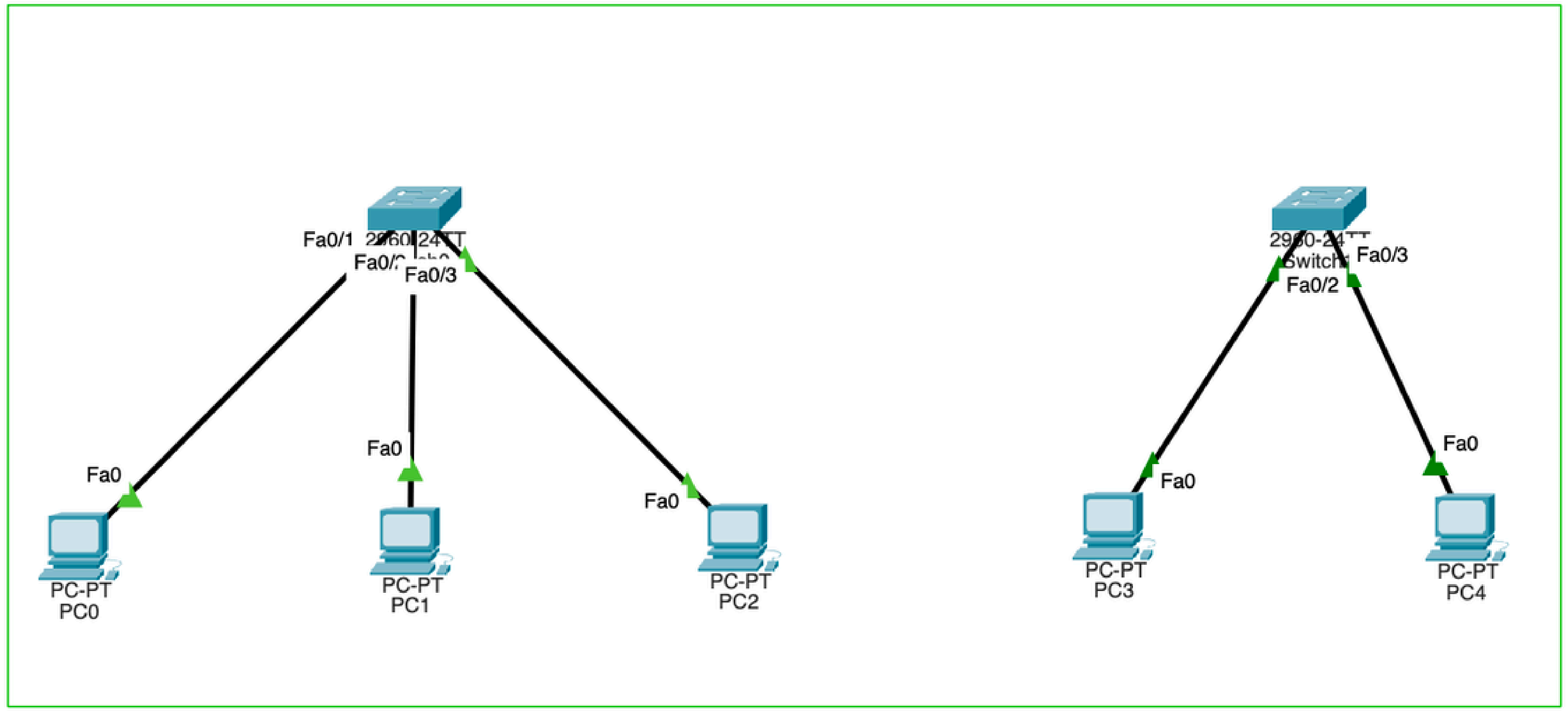
VLAN!

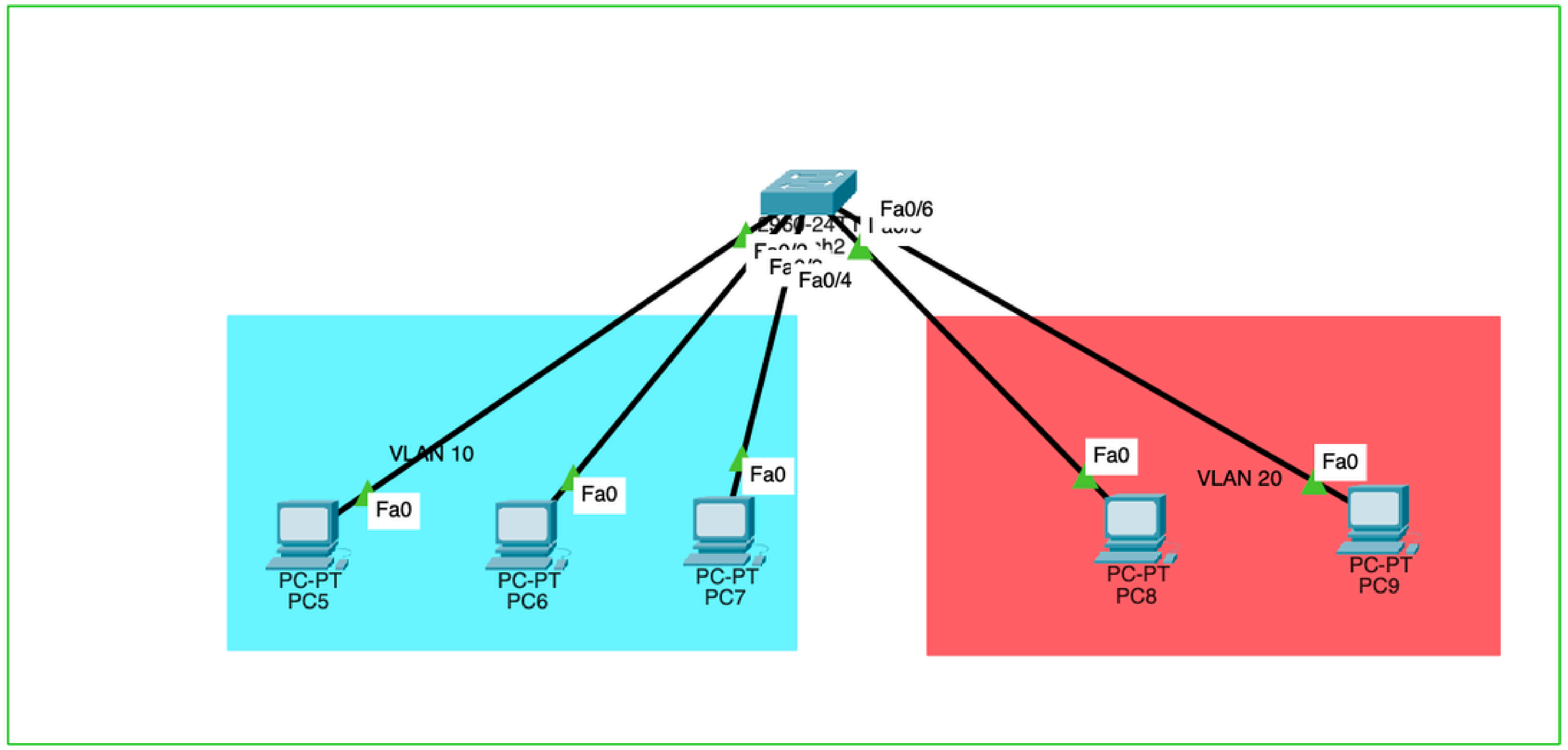
VLAN (Virtual Local Area Network) is a logical network created **within a physical switch**

Devices in **different VLANs** cannot communicate directly - only through a **router** or Layer 3 switch

Splits one large broadcast domain into smaller ones → **LESS CONGESTION!**




Fa0/1
Fa0/3
Fa0
Fa0
Fa0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
Fa0/3
Fa0/2
Fa0
Fa0
PC-PT
PC3
PC-PT
PC4


Fa0/6
Fa0/4
VLAN 10
Fa0
Fa0
Fa0
PC-PT
PC5
PC-PT
PC6
PC-PT
PC7
Fa0
VLAN 20
Fa0
PC-PT
PC8
PC-PT
PC9

VLAN tagging